

# PACE-FLU-SEV: A Bayesian Phase-Adaptive Connectivity and Epidemic-Memory Framework for Probabilistic Influenza Forecasting in Nigeria

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## Abstract

For the MAEPiMS Challenge 2026, we were required to develop an original forecasting approach based solely on the supplied simulated Nigerian influenza surveillance data. We developed PACE-FLU-SEV as a Bayesian framework that first forecasts weekly cases for Nigeria’s 36 states and the Federal Capital Territory and then propagates these trajectories to hospitalizations and deaths. To forecast cases, the core PACE-FLU-SEV model brings together four components: an eight-week epidemic-memory term, a population-weighted national signal, data-driven directed cross-state connectivity, and a continuous measure of epidemic phase. The model includes a phase-by-connectivity interaction, allowing the predictive contribution of cross-state information to depend on where the target state is in its epidemic phase. Hospitalizations and deaths are modeled through a linked Bayesian severity cascade, with uncertainty propagated draw-by-draw from cases to hospitalizations and deaths.

We evaluated the developed model using leakage-free rolling-origin validation across three simulated seasons, covering 24 one-week-ahead forecast origins and 888 state-origin forecasts. Across the one-week-ahead forecasts, mean absolute error was 2,446.75 cases, mean weighted interval score (WIS) was 1,365.31, and mean continuous ranked probability score (CRPS) was 1,705.82. The corresponding empirical coverage rates for the 50%, 80%, and 90% intervals were 49.2%, 76.9%, and 88.4%. The phase-by-connectivity interaction was positive at all 24 forecast origins, with its 90% posterior credible interval lying entirely above zero at every origin. In the integrated one-step validation, we obtained MAEs of 2,448.60 for cases, 32.57 for hospitalizations, and 3.95 for deaths; the corresponding 90% coverage rates were 88.1%, 92.5%, and 97.0%. A separate recursive stress test across 18 origins showed that calibration deteriorates with increasing open-loop forecast horizon, supporting the use of weekly rolling updates in operational settings. We generated the final production forecast at week 40 of 2025/2026, forecasting weeks 41–52 using 2,000 coherent posterior trajectories. For the final fits, the case, hospitalization, and death models all met the prespecified MCMC convergence criteria.

## 1 Introduction

For epidemic planning, probabilistic forecasts are useful because they characterize not only the expected future burden but also the uncertainty associated with that expectation. The quality of probabilistic forecasts can be assessed using proper scoring rules such as CRPS and WIS, which reward predictive distributions that are well calibrated while remaining sharp [1, 2]. Forecasting influenza is difficult because epidemic patterns change over time and differ across locations. States may move through growth, peak, and decline at different times, so information from other locations may be more useful at some stages of the season than at others.

Bayesian hierarchical modelling, negative-binomial observations, temporal memory, spatial information sharing, and spatiotemporal interaction kernels are all established components of infectious-disease models. A Bayesian approach to spatiotemporal interactions was developed by [3] using a continuous

interaction kernel learned within the model. Dynamic geographic grouping has also been incorporated into recent respiratory-virus forecasting, with geo-hierarchical ensembles using hospitalization trends to group geographic units [4]. Instead, the specific contribution of PACE-FLU-SEV is to allow the *strength of data-driven cross-state predictive borrowing to vary explicitly with a continuous epidemic-phase signal* within a Bayesian count-forecasting model.

The data provided for the challenge cover three simulated seasons (2023/2024, 2024/2025, and 2025/2026) and include weekly surveillance information at both state and national levels, as well as population denominators, geographic zones, and severity outcomes. The forecasting task focuses primarily on weekly influenza cases, with hospitalizations and deaths included as secondary forecasting targets. The report details the assumptions and final architecture of PACE-FLU-SEV, presents the evidence from model validation and the final production forecast, examines forecast uncertainty, and discusses the framework’s limitations and operational implications.

## 2 Model Assumptions

The model is fitted exclusively using the data supplied for the challenge. We treat each state’s population as fixed throughout the study period. The temporal structure of the model is indexed according to epidemiological week-of-season. For the macro-regional summaries, we define North as the combination of NW, NE, and NC, and South as the combination of SW, SE, and SS. The connectivity learned from the data represents conditional predictive relationships between states; it is not intended to measure physical mobility or identify causal transmission pathways. Hospitalizations and deaths are modeled predictively, not as mechanistic clinical-progression processes.

The final production forecast uses observed data through week 40 of 2025/2026 and recursively simulates weeks 41–52. As a result, our full-season summaries for cumulative burden, attack rate, and the seasonal peak are based on observed data for weeks 1–40 combined with simulated trajectories for weeks 41–52. A pre-week-40 peak belongs to the observed portion of the season rather than the forecast period, so it should be reported as an already-observed peak rather than as a prospective model prediction.

## 3 Forecasting Methodology

### 3.1 Population-standardized epidemic signal

For state  $i$  and week  $t$ , with cases  $Y_{it}$  and population  $P_i$ , is defined as

$$X_{it} = \log \left( 1 + 100000 \frac{Y_{it}}{P_i} \right). \quad (1)$$

### 3.2 Epidemic memory

The within-state history is summarized by an eight-week exponentially weighted memory,

$$M_{it} = \sum_{\ell=0}^7 K_{\ell} X_{i,t-\ell}, \quad K_{\ell} = \frac{\exp(-0.35\ell)}{\sum_{q=0}^7 \exp(-0.35q)}. \quad (2)$$

This provides weight to current incidence while retaining information from the preceding seven weeks.

### 3.3 Continuous epidemic phase

We defined growth as

$$G_{it} = X_{it} - X_{i,t-1}, \quad (3)$$

and the soft growth-phase quantity is

$$\pi_{it} = \text{logit}^{-1}(G_{it}/s_G), \quad (4)$$

where  $s_G$  is estimated from training data only at each forecast origin. This avoids unstable hard classification around epidemic transitions.

### 3.4 Predictive cross-state connectivity

For each directed target-source pair  $(i, j)$ , the training data are used to fit a regression of next-week target log incidence on target memory, national activity, target phase, and current source-state log incidence. If at least 20 training observations are available, the fitted source coefficient  $\hat{\gamma}_{ij}$  is retained when positive. Target-specific weights are normalized:

$$W_{ij} = \frac{\max(\hat{\gamma}_{ij}, 0)}{\sum_{k \neq i} \max(\hat{\gamma}_{ik}, 0)}, \quad j \neq i. \quad (5)$$

If no positive estimable edge exists for a target, equal cross-state weights are used. The borrowing signal is

$$B_{it} = \sum_{j \neq i} W_{ij} X_{jt}. \quad (6)$$

Connectivity is re-estimated using only information available at each validation origin. At the final week-40 production origin, 1,332 directed edges were estimable, 835 had positive coefficients, and no state required the equal-weight fallback.

### 3.5 National epidemic signal

The national signal is population weighted:

$$N_t = \log \left( 1 + 100000 \frac{\sum_i Y_{it}}{\sum_i P_i} \right). \quad (7)$$

## 4 Model Architecture

### 4.1 Case model: PACE-FLU-SEV

Weekly influenza cases were modeled using a hierarchical negative-binomial distribution and the model is defined as

$$Y_{i,t+1} \sim \text{NegativeBinomial}(\mu_{i,t+1}, \phi_C) \quad (8)$$

with

$$\begin{aligned} \log \mu_{i,t+1} = & \log(P_i/100000) + \alpha + \alpha_i + \beta_M M_{it} + \beta_N N_t \\ & + \beta_B B_{it} + \beta_P \pi_{it} + \beta_{BP} B_{it} \pi_{it}. \end{aligned} \quad (9)$$

State heterogeneity is represented by  $\alpha_i = \sigma_\alpha z_i$ ,  $z_i \sim N(0, 1)$ . The defining PACE-FLU-SEV term is  $\beta_{BP} B_{it} \pi_{it}$ , which modifies the contribution of cross-state borrowing according to epidemic phase.

## 4.2 Hospitalization layer

Hospitalizations were modeled as

$$H_{i,t+1} \sim \text{NegativeBinomial}(\mu_{i,t+1}^H, \phi_H), \quad (10)$$

$$\log \mu_{i,t+1}^H = \alpha_H + \alpha_i^H + \beta_C^H \log(1 + C_{i,t+1}) + \beta_{Hm}^H \log(1 + H_{it}) + \beta_P^H \pi_{it}. \quad (11)$$

## 4.3 Mortality layer

Deaths cases were modeled as

$$D_{i,t+1} \sim \text{NegativeBinomial}(\mu_{i,t+1}^D, \phi_D), \quad (12)$$

$$\log \mu_{i,t+1}^D = \alpha_D + \alpha_i^D + \beta_C^D \log(1 + C_{i,t+1}) + \beta_H^D \log(1 + H_{i,t+1}) \quad (13)$$

$$+ \beta_{Dm}^D \log(1 + D_{it}) + \beta_P^D \pi_{it}. \quad (14)$$

We did not impose a strict same-week binomial constraint from hospitalizations to deaths because the supplied data include weeks in which deaths exceed hospitalizations in the same week. In the recursive forecast, we carry each case trajectory forward to the corresponding hospitalization trajectory and then use that trajectory to generate the corresponding death trajectory. This structure propagates uncertainty coherently across the forecasting cascade rather than relying on point forecasts as inputs.

## 4.4 Prior distributions and Bayesian estimation

We used weakly informative priors to regularize estimation while allowing the data to determine the main predictive relationships. For the case model, the overall intercept was assigned

$$\alpha \sim \mathcal{N}(0, 2^2),$$

while the coefficients for epidemic memory, national activity, cross-state borrowing, epidemic phase, and the phase-by-borrowing interaction were assigned

$$\beta_M, \beta_N, \beta_B, \beta_P, \beta_{BP} \sim \mathcal{N}(0, 1).$$

State-specific effects were modeled hierarchically as

$$\alpha_i = \sigma_\alpha z_i, \quad z_i \sim \mathcal{N}(0, 1), \quad \sigma_\alpha \sim \mathcal{N}^+(0, 1).$$

The case-model dispersion parameter was specified as

$$\phi_C = 0.05 + \exp(\eta_C), \quad \eta_C \sim \mathcal{N}(\log 5, 0.5^2).$$

For the hospitalization and mortality models, the intercepts were assigned

$$\alpha_H \sim \mathcal{N}(-4, 2^2), \quad \alpha_D \sim \mathcal{N}(-5, 2^2).$$

The coefficients relating cases to the corresponding severity outcome were assigned  $\mathcal{N}(1, 0.5^2)$  priors, while the remaining regression coefficients were assigned  $\mathcal{N}(0, 1)$  priors. State-specific effects followed the same hierarchical normal structure used for the case model. For both severity models, dispersion was specified as

$$\phi = 0.05 + \exp(\eta), \quad \eta \sim \mathcal{N}(\log 5, 0.75^2).$$

Posterior inference was performed in Stan using four Markov chains. Convergence was assessed using  $\hat{R}$ , bulk and tail effective sample sizes, divergent transitions, and maximum tree-depth diagnostics.

## 5 Model Justification and Originality

The choice of PACE-FLU-SEV was guided by leakage-free model development, with model selection extending beyond measures of in-sample fit. We chose the final case model after evaluating it against alternatives that were both simpler and more flexible. We also considered extensions in which the mean and dispersion varied with epidemic intensity. Although these models captured additional structure in the data, they did not consistently perform better in out-of-sample forecasts. The final model was therefore specified to retain sufficient flexibility while maintaining estimability, adequate calibration, and computational stability.

Our claim of originality is confined to a specific aspect of the PACE-FLU-SEV formulation. Previous work has used Bayesian spatiotemporal interaction models to learn spatial-temporal influence structures [3], and dynamic approaches to geographic grouping have also been applied to respiratory-virus forecasting [4]. To our knowledge, PACE-FLU-SEV is distinguished by explicitly interacting learned cross-location borrowing with continuous local epidemic phase, allowing cross-state predictive contributions to vary smoothly with phase. Although the severity cascade broadens the framework to include hospitalization and mortality forecasts, it is not presented as the principal methodological innovation of PACE-FLU-SEV.

## 6 Validation Design

### 6.1 One-week-ahead case validation

For each season, we evaluated the final PACE-FLU-SEV case model at weeks 20, 24, 28, 32, 36, 40, 44, and 48. At each origin  $t$ , all previous-season data were available, while current-season training predictors were restricted to weeks before  $t$ . The one-step-ahead design used information available through week  $t$  solely to predict the outcome in week  $t+1$ . The growth scale and cross-state connectivity were re-estimated from the data available at each origin. This leakage-free procedure produced 888 state-level forecasts across the three seasons.

### 6.2 Integrated severity validation

To maintain a consistent evaluation design across outcomes, we assessed the case, hospitalization, and mortality layers at the same 24 forecast origins. At each origin, the severity models were trained only on outcomes available at that time, with forecast uncertainty propagated from the case posterior to hospitalizations and subsequently to deaths.

### 6.3 Recursive multi-horizon stress test

We conducted a separate open-loop recursive experiment using six origins in each season—weeks 20, 24, 28, 32, 36, and 40—and forecast horizons of up to 12 weeks, producing 7,992 state forecasts. This analysis was designed to stress-test unupdated recursive forecasting rather than serve as the primary operational validation. The intended operational approach is weekly updating as new surveillance data arrive.

## 7 Results

### 7.1 One-week-ahead case performance

Table 1: One-week-ahead validation performance of PACE-FLU-SEV across 888 state-level forecasts.).

| Measure               | Value    |
|-----------------------|----------|
| MAE                   | 2,446.75 |
| Median absolute error | 289.50   |
| RMSE                  | 8,577.97 |
| Mean WIS              | 1,365.31 |
| Median WIS            | 172.55   |
| Mean CRPS             | 1,705.82 |
| Median CRPS           | 213.37   |
| Mean bias             | −388.05  |
| Median bias           | 40.75    |
| 50% interval coverage | 49.2%    |
| 80% interval coverage | 76.9%    |
| 90% interval coverage | 88.4%    |

Performance was heterogeneous by season. From Table 1, the MAE was approximately 1,449 in 2023/2024, 4,400 in the high-severity 2024/2025 season, and 1,491 in 2025/2026. Corresponding 90% coverage was 92.6%, 79.4%, and 93.2%. The highest epidemic-intensity quartile was also the most difficult, with 90% coverage falling to approximately 77.5%.

### 7.2 Evidence for phase-adaptive connectivity

At all 24 validation origins, the posterior median of  $\beta_{BP}$  was positive and its 90% credible interval excluded zero. Across the 24 origins, the mean posterior median was 0.334 and the median posterior median was 0.322, with origin-specific posterior medians ranging from 0.214 to 0.460. The posterior probability that  $\beta_{BP}$  exceeded zero was consistently high, with a mean  $P(\beta_{BP} > 0)$  of approximately 0.9996 across origins. A similar pattern was observed for the main borrowing coefficient, which was positive at each of the 24 validation origins. These results support the central predictive hypothesis in the challenge data without implying causal interstate transmission.

### 7.3 Integrated case-hospitalization-death validation

Table 2: Integrated one-week-ahead validation across 24 origins.

| Outcome          | MAE      | RMSE     | Mean WIS | 90% coverage |
|------------------|----------|----------|----------|--------------|
| Cases            | 2,448.60 | 8,472.86 | 1,362.61 | 88.1%        |
| Hospitalizations | 32.57    | 105.76   | 18.90    | 92.5%        |
| Deaths           | 3.95     | 12.99    | 2.24     | 97.0%        |

Table 2 shows that calibration differed across outcomes and seasons. The high-severity 2024/2025 season was particularly challenging for cases and hospitalizations, while mortality intervals were comparatively conservative. In contrast, the predictive intervals for mortality remained conservative.

## 7.4 Recursive horizon sensitivity

Based on Figure 1, across the 18-origin recursive stress test, with forecasts extending up to 12 weeks ahead, we obtained an overall MAE of 6,089.06, a mean WIS of 4,879.16, and 90% coverage of 47.7%. One-week-ahead coverage was 85.0%, with substantial deterioration at longer horizons. Several factors contribute to the longer-horizon degradation, including the accumulation of trajectory uncertainty, propagation of turning-point errors, and inadequate aggregate dispersion as open-loop recursion continues. This finding shows why long recursive trajectories should not remain unchanged as new data become available. In operational use, the model should instead be re-estimated on a rolling basis rather than relying on the original week-40 forecast many weeks later.

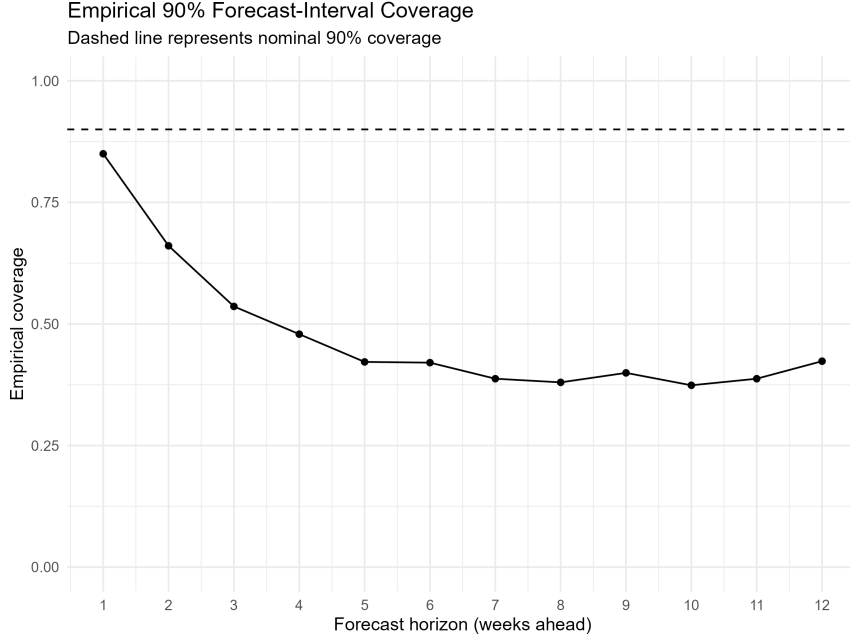


Figure 1: Observed 90% interval coverage by forecast horizon in the recursive multi-origin stress test. The dashed line marks nominal 90% coverage.

## 7.5 Final production fit and forecast

The final week-40 production fit used 2,000 posterior trajectories and all 37 states/FCT. All three components passed the prespecified computational checks (Table 3).

Table 3: Final production-run MCMC diagnostics.

| Model            | Max $\hat{R}$ | Min bulk ESS | Min tail ESS | Div. | Tree hits |
|------------------|---------------|--------------|--------------|------|-----------|
| Cases            | 1.0035        | 1104         | 1689         | 0    | 0         |
| Hospitalizations | 1.0032        | 2212         | 4071         | 0    | 0         |
| Deaths           | 1.0067        | 791          | 1362         | 0    | 0         |

The final posterior estimates for the case model had medians of 0.0540,  $-0.1453$ , 0.8925, 0.4139, and 0.3271 for  $\beta_M$ ,  $\beta_N$ ,  $\beta_B$ ,  $\beta_P$ , and  $\beta_{BP}$ , respectively. All posterior samples of  $\beta_{BP}$  were positive, and its approximate 90% credible interval ranged from 0.2883 to 0.3677. The negative national coefficient represents a conditional adjustment after local memory and cross-state borrowing, not a protective biological effect.

For the hospitalization model, we obtained posterior medians of 0.7602 for predicted cases, 0.2154 for hospitalization memory, and 0.6054 for phase. Posterior medians in the mortality model were 0.5220

for predicted cases, 0.3714 for predicted hospitalizations, 0.1044 for death memory, and 0.2898 for phase. The estimated coefficients describe predictive relationships on the scales used in the models and do not provide evidence of causal effects.

Across the forecast period, the national median weekly case forecast increases from 27,977 cases at week 41 to 46,471 cases at week 52. The corresponding median forecasts also increase for the severity outcomes, with hospitalizations rising from 440 to 762 and deaths from 45 to 73. The predictive intervals widen as the recursive horizon increases, reflecting greater uncertainty in forecasts further into the future.

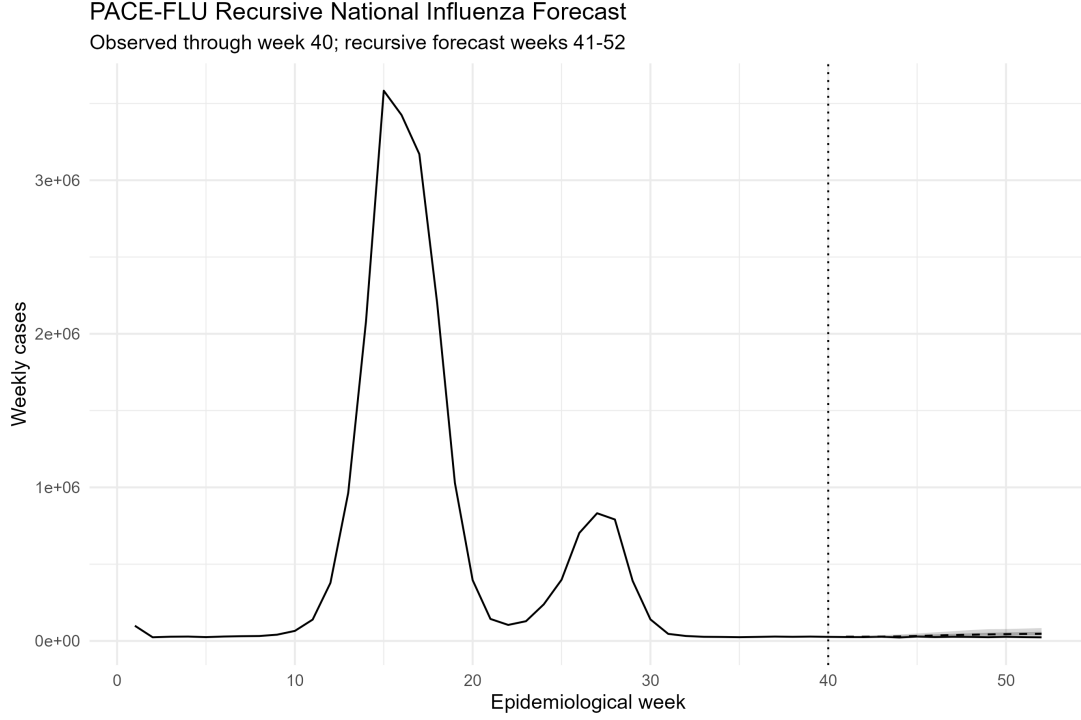


Figure 2: National 2025/2026 cases. Observations are shown for the full supplied season for retrospective assessment; the production forecast was generated using observations only through week 40. Shaded regions show 50% and 90% posterior predictive intervals for weeks 41–52.

The full-season national medians were 22,376,645 cumulative cases, 330,190 hospitalizations, 30,860 deaths, and a 9.7% case attack rate, based on observed weeks 1–40 and simulated weeks 41–52. Week 15 was identified as the national peak in all production trajectories because it falls within the observed portion of the season and had already occurred before forecasting began at week 40. Accordingly, we treat week 15 as part of the completed-season summary, not as a peak week that the model predicted prospectively.

## 8 Uncertainty Analysis

At each stage of recursive forecasting, PACE-FLU-SEV retains uncertainty in the posterior parameters together with predictive variation generated by the negative-binomial model. We first aggregate geographic totals within each posterior predictive trajectory and then calculate the forecast quantiles, instead of summing marginal quantiles from individual states. The resulting probabilistic forecasts report the 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, and 0.95 quantiles for cases, hospitalizations, and deaths at the state, geopolitical-zone, North/South, and national levels. Complete forecasts are available in `PACE_FLU_SEV_WEEKLY_SUBMISSION.csv` and `PACE_FLU_SEV_QUANTILE_SUBMISSION.csv`.

According to the final checks in `08_SUBMISSION_QC.csv`, the submission contained 1,656 wide-format

forecast rows and 11,592 long-format quantile rows. The submission passed the checks for completeness and quantile validity: no count quantiles were missing or negative, their ordering was correct, and all required weeks (41–52), three outcomes, 37 states/FCT, six geopolitical zones, and two macro-regions were represented. Convergence diagnostics were evaluated independently of the submission quality checks and are reported in `01_FINAL_MODEL_DIAGNOSTICS.csv`; all final model components satisfied the specified convergence criterion.

Calibration of the predictive uncertainty differed across the settings examined in the validation analyses. We observed substantially better calibration at one week ahead than over longer open-loop recursive horizons, with forecasting remaining more difficult during periods of high epidemic intensity. The differences in calibration across forecast settings are documented in the outputs generated by the one-step and multi-horizon validation analyses. The reported intervals quantify model-based predictive uncertainty conditional on the supplied synthetic surveillance process, not guaranteed nominal coverage for every state and forecast week.

## 9 Policy and Operational Implications

The framework is designed as a surveillance-support tool and is not intended to translate forecasts directly into automatic intervention decisions. At each location and forecast week, the median summarizes the central forecast, while the predictive intervals characterize the surrounding uncertainty, allowing planners to consider both when reviewing expected burden. The model also estimates state-level probabilities of increasing, decreasing, or unchanged burden. These outputs can help surveillance teams review data quality, assess preparedness, plan hospital capacity, and identify areas that may require greater surveillance attention.

The multi-horizon validation supports an important operational principle: forecasts should be updated as new weekly surveillance data become available. In practice, the system is intended to operate through repeated forecast updates rather than by issuing one 12-week trajectory and using it unchanged throughout the forecast period. The connectivity weights represent predictive associations and should not be interpreted as measures of travel flows or as evidence for assigning causal responsibility to individual states.

## 10 Discussion

The PACE-FLU-SEV framework combines information from local epidemic history, national influenza activity, learned cross-state predictive connectivity, and epidemic phase, while propagating forecasts through the severity outcomes. The strongest evidence supporting the core model design is the consistently positive phase-by-connectivity interaction across all 24 one-step validation origins. This indicates that the predictive value of cross-state information varies with epidemic phase rather than contributing a fixed spatial effect.

The framework also illustrates the distinction between model novelty and model completeness. Negative-binomial Bayesian count models, spatiotemporal interactions, dynamic geographic grouping, and probabilistic forecast scoring are established ideas [1–4]. The contribution here is their specific phase-adaptive predictive-borrowing formulation and its leakage-free empirical validation. The hospitalization and mortality layers broaden the target set while preserving coherent uncertainty propagation.

The principal weakness is long-horizon calibration. Recursive errors accumulate, abrupt turning points are difficult to anticipate, and aggregation exposes missing shared future shocks. These findings argue against post-hoc overconfidence and against presenting the 12-week open-loop trajectory as equally reliable at every horizon. They also explain why more elaborate model-development variants were not retained solely because they added flexibility.

## 11 Limitations

## 12 Limitations

The study has several limitations. First, the challenge data are simulated, so the forecasting performance observed here does not establish real-world effectiveness in Nigeria. Second, only three synthetic influenza seasons were available, and rolling forecast origins within the same season should not be regarded as independent epidemics. Third, the hospitalization and mortality components are predictive rather than mechanistic models of clinical disease progression. Fourth, calibration deteriorated at longer recursive forecast horizons, particularly around epidemic turning points. Finally, seasonal peak summaries generated from a late forecast origin may be dominated by a peak that has already occurred.

## 13 Conclusions

We developed PACE-FLU-SEV as a reproducible Bayesian framework for forecasting influenza activity in the synthetic Nigerian data provided by MAEPiMS. Across the validation periods, we found that the final case model was well calibrated for one-week-ahead probabilistic forecasts. Across all 24 leakage-free forecast origins, the phase-by-connectivity interaction remained positive, supporting the model’s central predictive hypothesis that the contribution of cross-state information changes with epidemic phase. We also observed good predictive performance from the severity components for both hospitalizations and deaths. The model fitted at the week-40 forecast origin satisfied the specified convergence criteria, and the resulting forecast outputs passed all submission-quality checks. The final forecasts quantify uncertainty in cases, hospitalizations, and deaths across geographic levels, from individual states to the national level, through week 52.

The evidence supports weekly rolling updates, with additional caution during high-intensity transitions and at longer forecast horizons. Interpretation of the framework requires two cautions: the cross-state connectivity describes predictive, not causal, relationships, and the posterior predictive uncertainty should be considered in full rather than reducing each forecast to its median.

## Reproducibility and Source Code

## 14 Limitations

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**GitHub repository:** <https://github.com/drtaiwo/PACE-FLU-SEV-MAEPiMS-Challenge->

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